

DAMG 7370 – Final Project Report

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Toronto Bicycle Thefts Data Analysis

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1 Dataset Introduction

1.1 Background & Motivation

Bicycle theft is a persistent urban challenge in Toronto. While the Toronto Police Service publishes detailed theft records as open data, the raw CSV format containing over 39,000 entries is not directly actionable for decision-makers. Cyclists cannot assess when and where they face highest risk. City planners lack empirical evidence for resource allocation. Law enforcement struggles to identify spatial and temporal patterns for targeted interventions.

This project addresses these limitations by building a cloud-based analytics pipeline that transforms raw police incident data into interactive, user-friendly dashboards. The solution enables three key stakeholder groups to make data-driven decisions: cyclists understanding personal risk profiles, city planners allocating prevention resources strategically, and law enforcement identifying hotspots and high-risk periods for patrol optimization.

1.2 Dataset Overview

The Toronto Bicycle Thefts dataset from the City of Toronto Open Data Portal contains granular, event-level theft incidents spanning 10 years (2014-2024). The dataset structure includes:

- **Event information** – Unique event ID, primary offence classification, occurrence and report dates, plus derived temporal attributes (year, month, day of week, hour)
- **Location context** – Police division assignment, general location type (apartment, street, commercial), specific premises type (house, parking lot, transit station)
- **Bike attributes** – Make, model, type, speed, color, estimated cost, and current status (STOLEN, RECOVERED, UNKNOWN)
- **Geospatial data** – Longitude and latitude coordinates in WGS84 projection system for mapping

With over 39,000 time-stamped records across multiple analytical dimensions, this dataset is well-suited for temporal trend analysis, geospatial hotspot identification, and multi-variate pattern recognition. The comprehensive 10-year timespan enables longitudinal analysis of theft patterns, policy intervention effectiveness, and emerging trends.

2 Objectives

This project performs comprehensive crime pattern analysis for Toronto bicycle thefts through an end-to-end cloud analytics pipeline. The primary objectives are:

1. **Pipeline Implementation:** Design and deploy a Kafka-based ETL pipeline on AWS that ingests raw bicycle theft CSV data, performs streaming-style processing, cleans and standardizes records, and stores curated datasets in both Amazon S3 (data lake) and Amazon RDS MySQL (relational analytics layer)
2. **Temporal Analysis:** Identify high-risk time windows across multiple scales - yearly trends, seasonal patterns, day-of-week distributions, and hourly concentrations - to understand when thefts are most likely to occur
3. **Spatial Analysis:** Analyze geographic theft concentration by police division, neighborhood, premises type, and location category to pinpoint high-vulnerability areas

4. **Attribute Analysis:** Explore bike characteristics including brand, type, color, and cost to understand targeting patterns and asset selection by thieves
5. **Recovery Pattern Analysis:** Examine theft outcomes across divisions and contextual factors to assess police performance and identify what aids successful recovery
6. **Interactive Visualization:** Deliver Amazon QuickSight dashboard with global filters, drill-down capabilities, and cross-filtering that enables non-technical stakeholders to explore multi-dimensional patterns

2.1 Team Responsibilities & Scope

Team roles are clearly divided: Chaitanya (cloud infrastructure, Apache Kafka setup, Python ETL development), Mudra (MySQL schema design, SQL query development, analytical views creation), Qi (QuickSight configuration, dashboard design, visual storytelling).

Project scope focuses on batch-style analytics and visualization, **excluding** real-time alerting systems, production-grade security hardening, continuous integration/deployment pipelines, cost optimization analysis, and advanced machine learning predictive models. The deliverable is a functional prototype demonstrating cloud-native data engineering and business intelligence best practices.

3 Data Architecture

The solution architecture is deployed entirely on Amazon Web Services, implementing a streaming-plus-batch ETL pattern optimized for analytical workloads. The five core architectural components are:

- **Amazon EC2** – Hosts Apache Kafka broker, ZooKeeper coordination service, and Python producer/consumer ETL scripts on a t3.xlarge compute instance
- **Apache Kafka + ZooKeeper** – Provides distributed message streaming backbone between raw data source and ETL processing layer
- **Amazon S3** – Serves as scalable data lake storing cleaned bicycle theft dataset in CSV format for long-term retention and cross-team access
- **Amazon RDS (MySQL)** – Delivers relational analytics layer with optimized schema, indexed tables, and pre-calculated SQL views
- **Amazon QuickSight** – Cloud-native business intelligence platform providing interactive dashboard interface with in-memory SPICE engine

Data Flow: Toronto Open Data Portal (CSV download) → Amazon EC2 (Kafka producer streams messages → Kafka consumer applies pandas ETL) → Amazon S3 (cleaned CSV storage) → Amazon RDS MySQL (relational tables + analytical views) → Amazon QuickSight (interactive dashboards with global filters).

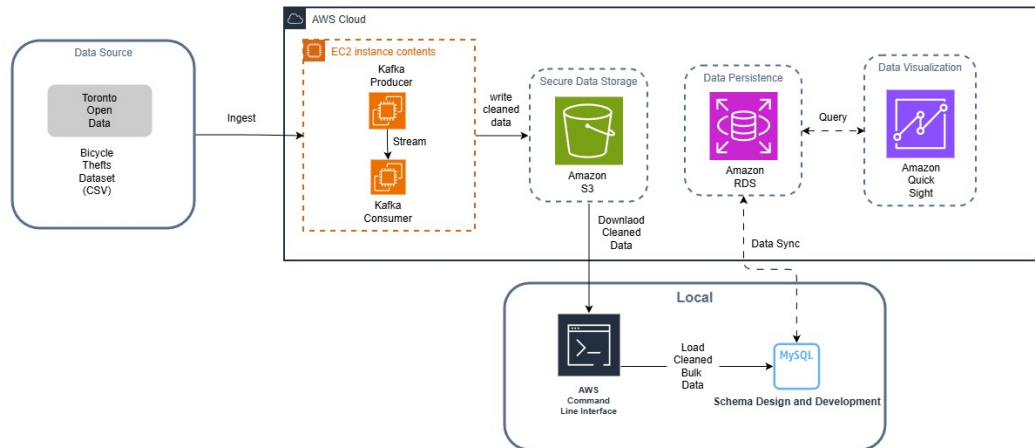


Figure 1: End-to-end pipeline architecture: data flows from Toronto Open Data through Kafka streaming to S3 storage, MySQL analytics, and QuickSight visualization

This architecture balances several design considerations: Kafka provides streaming abstraction even for static CSV (meeting course requirements), S3 enables data lake pattern with cross-account access, RDS MySQL supports complex analytical queries with sub-second performance, and QuickSight delivers interactive user experience without custom frontend development.

4 Pipeline Implementation – Infrastructure & Preprocessing

Chaitanya architected and implemented the complete cloud infrastructure and ETL pipeline, managing data flow from raw Toronto Open Data through cleaned analytical datasets.

4.1 Infrastructure Deployment

The pipeline runs on Amazon EC2 (t3.xlarge instance in us-east-1) hosting Apache Kafka, ZooKeeper, and Python ETL scripts. Instance configuration includes Java runtime installation (Kafka dependency), Kafka broker setup with public IP advertised listeners, JVM heap tuning (1GB for Kafka, 512MB for ZooKeeper), and security group rules allowing SSH and Kafka access. Python environment includes kafka-python, pandas, boto3, and python-dateutil libraries.

4.2 Streaming Architecture

A Python producer script reads the raw bicycle theft CSV using `csv.DictReader`, iterates through 39,000+ rows, and publishes each record as JSON message to Kafka topic `toronto_bicycle.thefts_project`. A Python consumer script subscribes to this topic, accumulates messages into pandas DataFrame, applies comprehensive ETL transformations, and outputs cleaned dataset.

4.3 Data Cleaning & Transformation Logic

The pandas-based ETL implements six transformation categories: (1) **Column standardization** - lowercase names with underscore separation, (2) **Date parsing** - safe parsing with NaT handling for invalid entries, (3) **Numeric conversion** - type coercion with NaN for non-numeric values, (4) **Text normalization** - uppercase conversion and whitespace trimming for consistent grouping, (5) **Null handling** - removal of rows missing critical fields (occurrence dates, geospatial coordinates), (6) **Duplicate elimination** - exact duplicate row removal preventing double-counting.

The cleaned dataset is persisted to both local EC2 storage and Amazon S3 (s3://toronto-bicycle-thefts/clean/Bicycle_Thefts_cleaned.csv) with cross-account access policies enabling team collaboration across separate AWS accounts.

5 Data Modeling & SQL Analysis

5.1 Database Configuration

Following the ETL pipeline completion, I established the relational analytics layer on Amazon RDS MySQL. The cleaned CSV from S3 was loaded into a table named `bicycle_thefts` with approximately 39,000 records spanning the full 10-year period.

Technical Challenge: Initial data loading attempts using MySQL Workbench's graphical Table Data Import Wizard encountered persistent timeout failures due to the large file size and row count. This was resolved by employing the `LOAD DATA INFILE` SQL command via terminal, which streams CSV data directly into the database table with significantly improved performance compared to the GUI's row-by-row processing approach.

5.2 Analytical Query Framework

I developed 15 production-quality SQL queries organized into four analytical categories, employing diverse techniques including aggregation, temporal grouping, window functions, conditional logic, and statistical calculations.

5.2.1 Category 1: Temporal Analysis (When Do Thefts Happen?)

Four queries examined theft patterns across time scales from annual to hourly granularity:

Query 1 - Annual Trends: Utilized `GROUP BY` with percentage calculations (`COUNT(*) * 100.0 / (SELECT COUNT(*) FROM bicycle.thefts)`) to analyze yearly distribution from 2014-2024

Query 2 - Monthly Seasonality: Aggregated by month with `AVG(bike_cost)` to identify seasonal patterns and value variations

Query 3 - Day of Week: Determined weekly patterns using day-of-week grouping and percentage analysis

Query 4 - Hourly Distribution: Applied `CASE` statements to categorize 24 hours into four time periods (Morning 6-11, Afternoon 12-17, Evening 18-21, Night 22-5)

Key Findings: Temporal analysis revealed bicycle thefts peaked in 2018 with 3,956 incidents (10.11% of total dataset) followed by sustained decline through 2024. The 2024 prelimi-

nary data shows 2,711 thefts representing a 9.06% decrease from 2023, indicating that prevention interventions and behavioral changes are having measurable positive impact. Seasonal analysis identified July as the highest-risk month with 5,996 thefts (15.3% of annual total), while winter months show 6x lower theft rates. Hourly analysis pinpointed 6pm as the most dangerous hour with 2,723 thefts, coinciding with evening rush hour when bikes are commonly left unattended at workplaces, transit stations, and commercial areas.

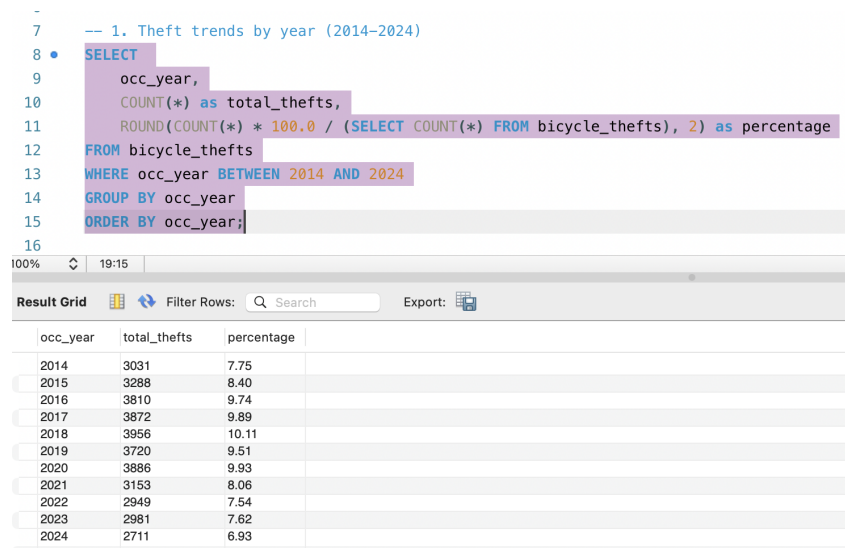


Figure 2: Annual theft trends 2014-2024: peak in 2018 followed by sustained multi-year decline

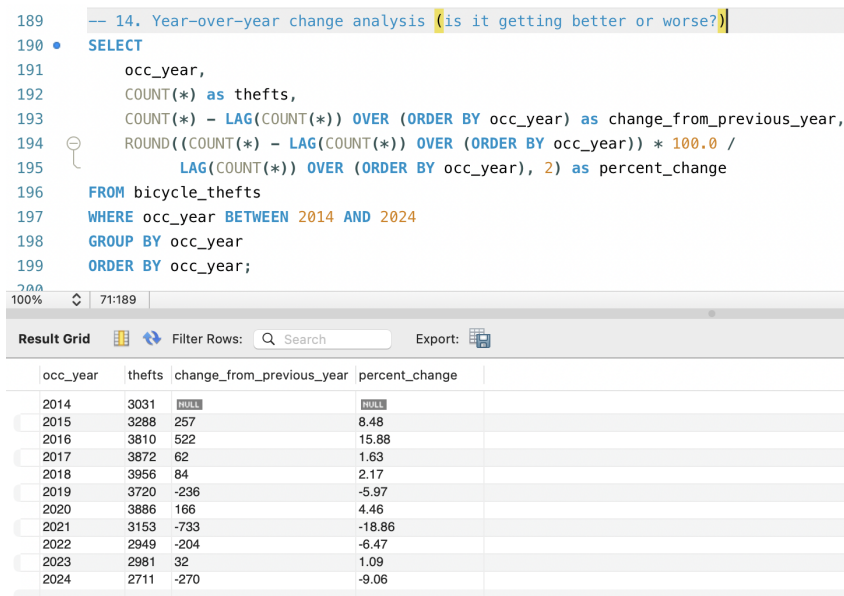


Figure 3: Year-over-year percentage changes using LAG() window function: growth period (2014-2018) and decline period (2019-2024) clearly delineated

5.2.2 Category 2: Geographic Analysis (Where Do Thefts Happen?)

Three queries examined spatial theft concentration and police performance:

Query 5 - Premises Hotspots: Analyzed theft distribution by premises type using conditional aggregation (COUNT(CASE WHEN status = 'RECOVERED' THEN 1 END)) for recovery rate

calculations

Query 6 - Location Categories: Examined broad location type patterns with average bike value analysis

Query 7 - Division Performance: Calculated total thefts, recovered cases, and recovery percentages across all 17 Toronto police divisions

Key Findings: Geographic analysis identified critical spatial concentration. Outside/Street locations account for 29.40% of all thefts (11,507 incidents), while apartments represent 24.45% (9,567 incidents). Combined, these two location types constitute 53.85% of all bicycle thefts in Toronto, indicating clear vulnerability patterns that can inform targeted prevention.

Most critically, recovery rate analysis revealed a systemic operational challenge. Across all 17 police divisions, not a single division achieves recovery rate exceeding 2%. The best-performing division (D31) recovers only 1.64% of stolen bikes, while the worst performer (D32) achieves 0.59%. This universally low recovery performance is independent of theft volume or geographic coverage, suggesting structural challenges in bike recovery processes rather than division-specific issues. The finding emphasizes that prevention is the only viable strategy, as stolen bicycles are statistically unlikely to be returned to owners.

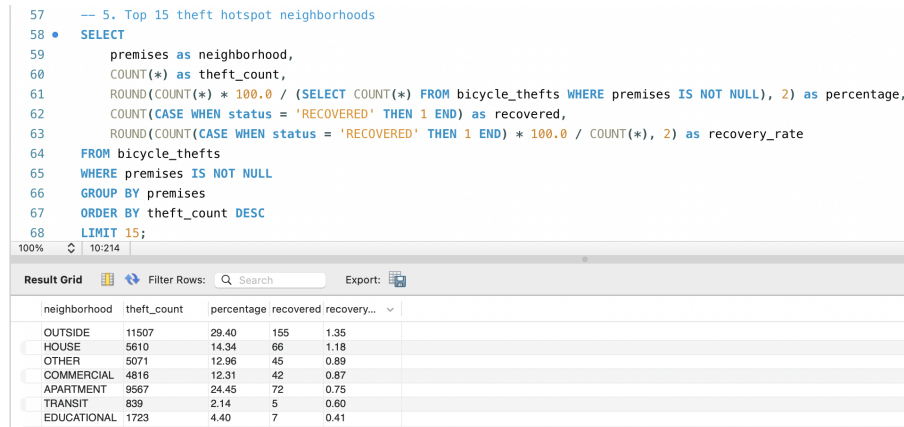


Figure 4: Theft distribution by premises type: Outside (29.40%) and Apartments (24.45%) represent majority of incidents

```

72     location_type,
73     COUNT(*) as theft_count,
74     ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM bicycle_thefts WHERE location_type IS NOT NULL), 2) as percentage,
75     ROUND(AVG(bike_cost), 2) as avg_bike_value
76 FROM bicycle_thefts

```

100% 47:75

Result Grid Filter Rows: Search Export:

location_type	theft_count	percentage	avg_bike_value
APARTMENT (ROOMING HOUSE, CONDO)	9567	24.45	1102.61
BANK AND OTHER FINANCIAL INSTITUTION...	258	0.66	1058.56
BAR / RESTAURANT	880	2.25	1153.68
COMMERCIAL DWELLING UNIT (HOTEL, MO...	119	0.30	1025.10
COMMUNITY GROUP HOME	8	0.02	519.71
CONSTRUCTION SITE (WAREHOUSE, TRAIL...	44	0.11	1281.84
CONVENIENCE STORES	372	0.95	998.15
DEALERSHIP (CAR, MOTORCYCLE, MARINE...	23	0.06	1611.10
GAS STATION (SELF, FULL, ATTACHED CON...	34	0.09	824.04
GO BUS	9	0.02	774.29
GO STATION	310	0.79	652.74
GO TRAIN	16	0.04	621.80
GROUP HOMES (NON-PROFIT, HALFWAY HO...	14	0.04	1067.40
GROUP HOMES (NON-PROFIT, HALFWAY HO...	36	0.09	876.12
HOMELESS SHELTER / MISSION	57	0.15	1485.55
HOSPITAL / INSTITUTIONS / MEDICAL FACIL...	411	1.05	1024.71
JAILS / DETENTION CENTRES	2	0.01	400.00
NURSING HOME	108	0.28	993.94
OPEN AREAS (LAKES, PARKS, RIVERS)	725	1.85	897.36
OTHER COMMERCIAL / CORPORATE PLACE...	3086	7.89	1201.78
OTHER NON COMMERCIAL / CORPORATE P...	501	1.28	1031.63
OTHER PASSENGER TRAIN	1	0.00	300.00
OTHER PASSENGER TRAIN STATION	8	0.02	706.25
OTHER REGIONAL TRANSIT SYSTEM VEHIC...	3	0.01	775.00
OTHER TRAIN ADMIN OR SUPPORT FACILITY	2	0.01	390.00
OTHER TRAIN TRACKS	1	0.00	NULL
OTHER TRAIN YARD	2	0.01	1300.00
PARKING LOTS (APT., COMMERCIAL OR NO...	3654	9.34	1141.96
PHARMACY	695	1.78	988.81
POLICE / COURTS (PAROLE BOARD, PROBA...	58	0.15	965.27
PRIVATE PROPERTY STRUCTURE (POOL, S...	2932	7.49	1184.30
RELIGIOUS FACILITIES (SYNAGOGUE, CHU...	57	0.15	1100.40

Figure 5: Granular location type breakdown across 25+ categories revealing apartment vulnerability and transit relative safety

5.2.3 Category 3: Bike Attribute Analysis (What Gets Stolen?)

Four queries examined bike characteristics and targeting patterns:

Query 8 - Brand Analysis: Identified most frequently stolen makes with recovery rate comparisons, filtering out empty strings and "UNKNOWN" entries

Query 9 - Color Patterns: Analyzed theft frequency by bike color using percentage calculations

Query 10 - Type Analysis: Examined mountain, road, electric, and other bike types with MIN/MAX/AVG cost analysis

Query 11 - High-Value Thefts: Focused on premium bikes exceeding \$1,000 using WHERE clause filtering

Key Findings: Brand analysis revealed that generic categories "OT" (Other) and "UK" represent 12,088 thefts, accounting for 30.8% of the total dataset. A critical data quality issue emerged: 945 bikes (2.4% of total) were labeled "UNKNOWN MAKE," indicating owners could not identify their bicycle manufacturer. This gap severely undermines recovery prospects, as police cannot match recovered bikes to theft reports without manufacturer identification.

Premium brand analysis shows marginal recovery advantages. Specialized bikes (average value \$1,880) achieve 2.97% recovery rate, while budget brands like UK (average value \$887) show only 0.26% recovery. However, all categories remain critically low, indicating that regardless of bike value or brand prestige, cyclists face similar statistical likelihood of permanent loss. This finding challenges the assumption that premium bikes warrant disproportionate security investment - all bikes require robust theft prevention measures.

```

97  -- 8. Most stolen bike makes (brands thieves target)
98  • SELECT
99      bike_make,
100      COUNT(*) as times_stolen,
101      ROUND(AVG(bike_cost), 2) as avg_value,
102      COUNT(CASE WHEN status = 'RECOVERED' THEN 1 END) as recovered,
103      ROUND(COUNT(CASE WHEN status = 'RECOVERED' THEN 1 END) * 100.0 / COUNT(*), 2) as recovery_rate
104  FROM bicycle_thefts
105  WHERE bike_make IS NOT NULL
106      AND bike_make != ''
107      AND bike_make != 'UNKNOWN'

```

bike_make	times_stolen	avg_value	recovered	recovery_rate
OT	7879	1021.76	51	0.65
UK	4209	887.57	11	0.26
GI	2267	982.58	14	0.62
OTHER	2172	1492.28	34	1.57
TR	2044	1008.26	8	0.39
NO	1275	822.87	6	0.47
UNKNOWN MAKE	945	966.70	12	1.27
CC	859	419.10	1	0.12
TREK	844	1545.46	16	1.90
SU	832	270.73	0	0.00
GIANT	804	1261.60	19	2.36
CA	782	1397.46	4	0.51
RA	728	579.93	6	0.82
SPECIALIZED	708	1880.00	21	2.97
SC	705	498.78	0	0.00

Figure 6: Brand targeting analysis: "OT" and "UK" dominate with 30.8% of thefts; all brands show 13% recovery rates

5.2.4 Category 4: Advanced Insights

Four complex queries provided multi-dimensional analysis:

Query 12 - Risk Matrix: Cross-tabulated time-of-day with location type for combined risk assessment using nested subqueries

Query 13 - Seasonal-Location Patterns: Analyzed seasonal theft patterns by premises type with HAVING clause filtering for statistical significance

Query 14 - Trend Analysis: Applied LAG() window function to calculate year-over-year absolute and percentage changes

Query 15 - Recovery Factors: Multi-dimensional analysis examining location and division combinations with minimum threshold filtering (HAVING total.thefts > 50)

Key Findings: Window function analysis precisely identified inflection points in Toronto's theft trajectory. The 2016 calendar year experienced the largest single-year increase (+15.88%, +522 thefts), while 2021 marked the sharpest decline (-18.86%, -733 thefts) correlating with COVID-19 pandemic behavioral changes including remote work adoption and reduced urban mobility. The sustained decline through 2024 (-9.06%) provides empirical evidence that prevention interventions, infrastructure improvements, or societal behavioral shifts are producing measurable positive impact.

5.3 Data Products for Visualization

To support Qi's QuickSight dashboard development, I architected two categories of data products:

5.3.1 Pre-Aggregated SQL Views

Six analytical views provide optimized query performance for common dashboard patterns:

1. `vw_hourly_thefts` - Hourly theft counts for time-of-day visualizations
2. `vw_top_locations` - Premises types with theft counts and recovery percentages for geographic mapping
3. `vw_yearly_trends` - Annual aggregations for time-series trend charts
4. `vw_monthly_patterns` - Monthly theft counts with average bike values for seasonal analysis
5. `vw_bike_brands` - Brand-level statistics with recovery metrics for target profiling
6. `vw_division_performance` - Police division benchmarking for operational assessment

These views reduce computational overhead by pre-calculating frequently accessed aggregations, enabling faster dashboard rendering and improved user experience.

5.3.2 Master Table Architecture

The critical innovation was `vw_master_table`, designed specifically for QuickSight's global filter functionality. This view incorporates all original columns plus three computed dimensions:

1. bike_make_clean (Brand Standardization): Raw data exhibited severe inconsistency in brand naming. For example, Giant bicycles appeared as "GI", "GIANT", "Giant Bicycle", "GIANT BICYCLE" - four variants referencing identical manufacturer. I implemented CASE logic consolidating variants for major brands (Giant, Trek, Norco, CCM, Supercycle, Raleigh, Specialized, Cannondale, Schwinn) while grouping generic entries (OT, OTHER, UNKNOWN MAKE, UK, ?, empty strings, NULL values) into single "UNKNOWN" category. This transformation collapsed 20+ naming variants into clean, analyzable categories.

2. season (Temporal Categorization): Created meteorological season classification mapping 12 calendar months to four seasons (Winter: Dec-Feb, Spring: Mar-May, Summer: Jun-Aug, Fall: Sep-Nov), enabling seasonal filtering revealing the 6x theft multiplier between summer and winter.

3. time_of_day (Hourly Grouping): Categorized 24-hour clock into four periods (Morning: 6-11, Afternoon: 12-16, Evening: 17-20, Night: 21-5), enabling time-of-day filtering that surfaces the evening rush hour concentration.

Architectural Impact: This master table design enables QuickSight's global filter paradigm where selecting "Summer + Evening + Apartments" instantly updates all dashboard visualizations (maps, charts, KPIs) across the entire 39,000+ record dataset. By preprocessing these dimensions at the database layer rather than calculating in the BI tool, we achieve sub-second dashboard response times even with complex multi-dimensional filtering.

5.4 Analysis Techniques

The SQL analysis employed multiple analytical methodologies:

- **Descriptive Analytics** - Aggregation (GROUP BY, COUNT) and frequency analysis understanding what patterns exist
- **Temporal Analysis** - Time-series pattern identification across year, month, day, hour granularities
- **Trend Analysis** - Window functions (LAG) calculating year-over-year changes and identifying inflection points

- **Comparative Analysis** - Recovery rate benchmarking across divisions and risk assessment across locations
- **Segmentation** - Multi-dimensional breakdowns identifying high-risk combinations of time, location, and attributes
- **Statistical Analysis** - Percentage calculations, averages (AVG), and distribution analysis

6 BI Implementation & Visualization Strategy

6.1 Technical Architecture

Qi implemented Amazon QuickSight as the business intelligence platform, leveraging several advanced capabilities for high-performance analytics:

SPICE Engine: QuickSight's Super-fast, Parallel, In-memory Calculation Engine caches the complete dataset in memory rather than executing live database queries. This architecture achieves sub-second load times for complex geospatial visualizations spanning 10 years and 39,000+ records, ensuring seamless user interactivity even with multiple simultaneous filter operations.

Master Table Pattern: Rather than connecting to multiple pre-aggregated views, QuickSight ingests the single `vw_master_table` containing granular records with enriched dimensions (`bike_make_clean`, `season`, `time_of_day`). This "flattened" approach enables dynamic on-the-fly aggregations, allowing users to slice data by any combination of attributes without pre-calculating every possible aggregation.

Geospatial Transformation: Raw latitude and longitude numeric columns were mapped to WGS84 Geospatial Data Types in QuickSight's metadata layer. This transformation enables the BI software to render coordinates as interactive map points with clustering algorithms rather than treating them as simple decimal numbers, unlocking spatial visualization capabilities.

6.2 Visualization Design

The dashboard implements four analytical lenses addressing "Where, When, and What" of bicycle theft:

6.2.1 Spatial Analysis (The "Where")

Interactive point map utilizes clustering algorithms to handle high-density data concentration. Conditional logic applies color encoding based on theft status (Orange = Stolen, Blue = Recovered), enabling users to visually distinguish neighborhoods with higher recovery success versus high-loss zones. Map interactivity supports zoom, pan, and point selection for detailed incident inspection.

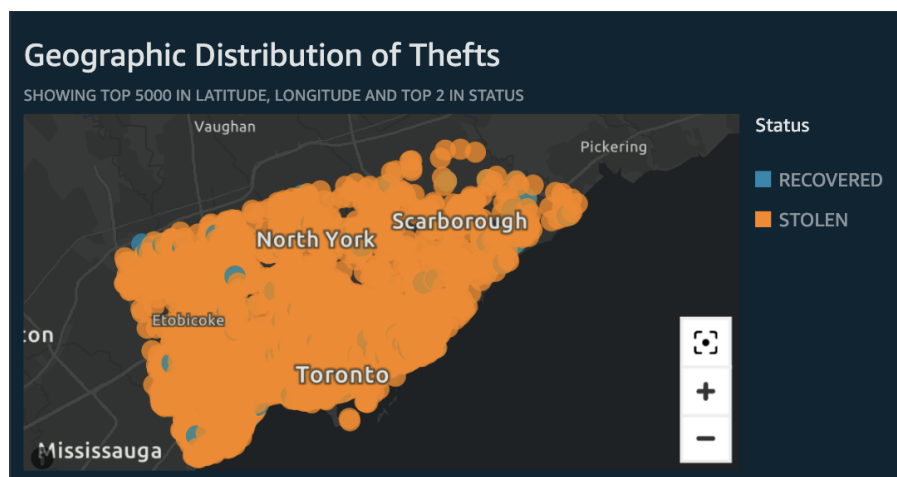


Figure 7: Interactive geospatial point map with clustering: visual encoding distinguishes stolen (orange) versus recovered (blue) incidents

6.2.2 Temporal Trend Analysis (The "When")

Time-series line chart tracks theft volume evolution across the 10-year dataset. A critical feature is the drill-down hierarchy on the X-axis: users can click annual data points to drill down to monthly granularity, revealing seasonal spikes obscured in aggregated annual views. This hierarchical navigation enables discovery of patterns at multiple temporal scales.

6.2.3 Behavioral Risk Analysis

Heat map matrix crosses day-of-week (Monday-Sunday) with time-of-day (Morning, Afternoon, Evening, Night) using color intensity to encode theft frequency. This two-dimensional visualization reveals behavioral patterns invisible in one-dimensional analyses, specifically identifying "danger zones" such as Friday evenings that represent disproportionately high theft risk.

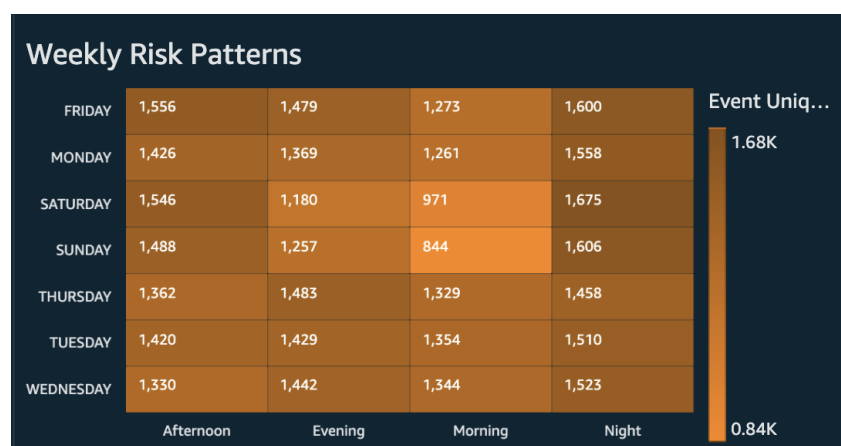


Figure 8: Day-of-week × time-of-day heatmap: color intensity reveals Friday evening danger zones

6.2.4 Target Analysis (The "What")

Bar chart visualization analyzes bike_make_clean attribute to identify most frequently stolen brands. A "Top 10" ranking filter focuses the view on highest-frequency targets while excluding

”Unknown” category to surface actionable brand-specific insights (Giant, Trek, Specialized targeting patterns).

6.3 Dashboard Architecture

The ”Toronto Bicycle Thefts Dashboard” implements ”Single Pane of Glass” design philosophy where the complete analytical story is visible without scrolling, regardless of screen size or device.

KPI Cockpit: Top-row metrics provide immediate situational awareness: Total Thefts (volume), Total Recovered (success count), Recovery Rate % (efficiency metric), Average Bike Value (economic impact). These KPIs dynamically recalculate based on active filter selections, providing context-aware insights.

Global Interactivity: Five global filter controls (Year, Neighborhood, Bike Type, Day of Week, Time of Day) enable multi-dimensional data slicing. Filters are linked across all visual components, ensuring synchronized updates throughout the dashboard.

Bi-Directional Filtering: Filter Actions configured on all charts enable cross-filtering workflows. For example, selecting a specific geographic location on the map automatically filters the heatmap and bar charts to display only data relevant to that location, facilitating focused geographic analysis.

Responsive Layout: Free-form layout with ”Fit to Window” scaling adapts to various screen sizes, ensuring optimal viewing experience across desktop monitors, laptops, and tablets without horizontal scrolling or layout breaking.

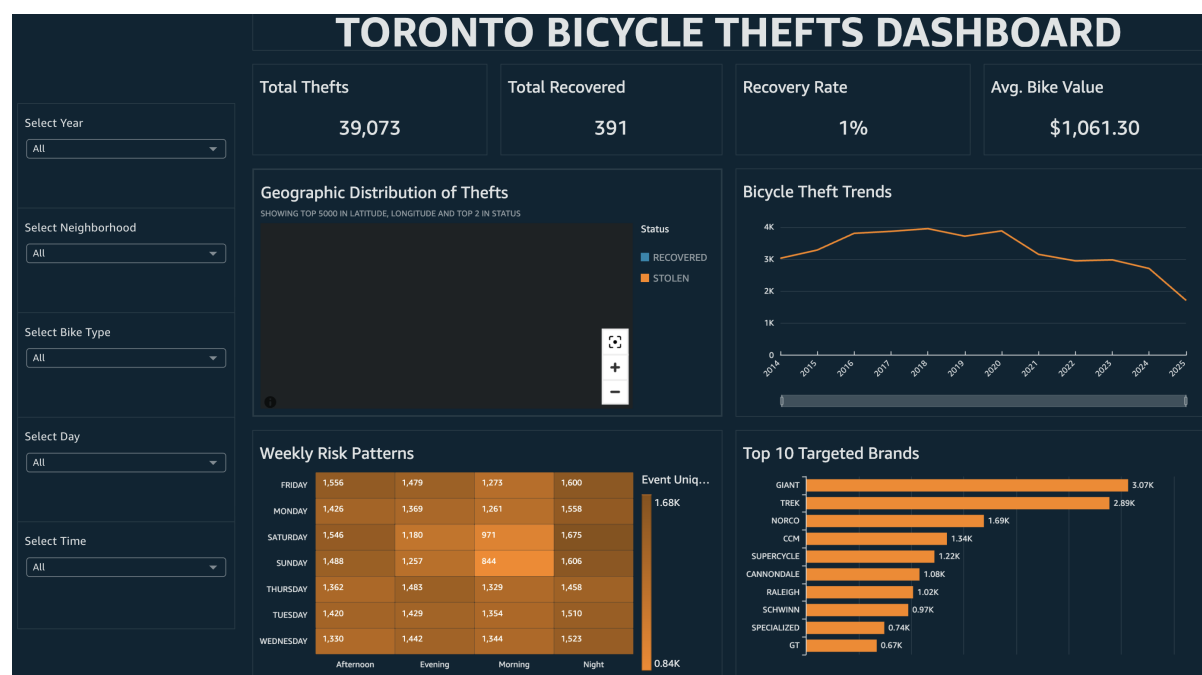


Figure 9: Complete QuickSight dashboard implementing single pane of glass design with KPI cockpit, global filters, and four-quadrant analytical layout

7 Comprehensive Risk Assessment

Synthesizing findings across temporal, geographic, and attribute analyses, we developed a comprehensive risk assessment matrix categorizing threat levels across multiple dimensions.

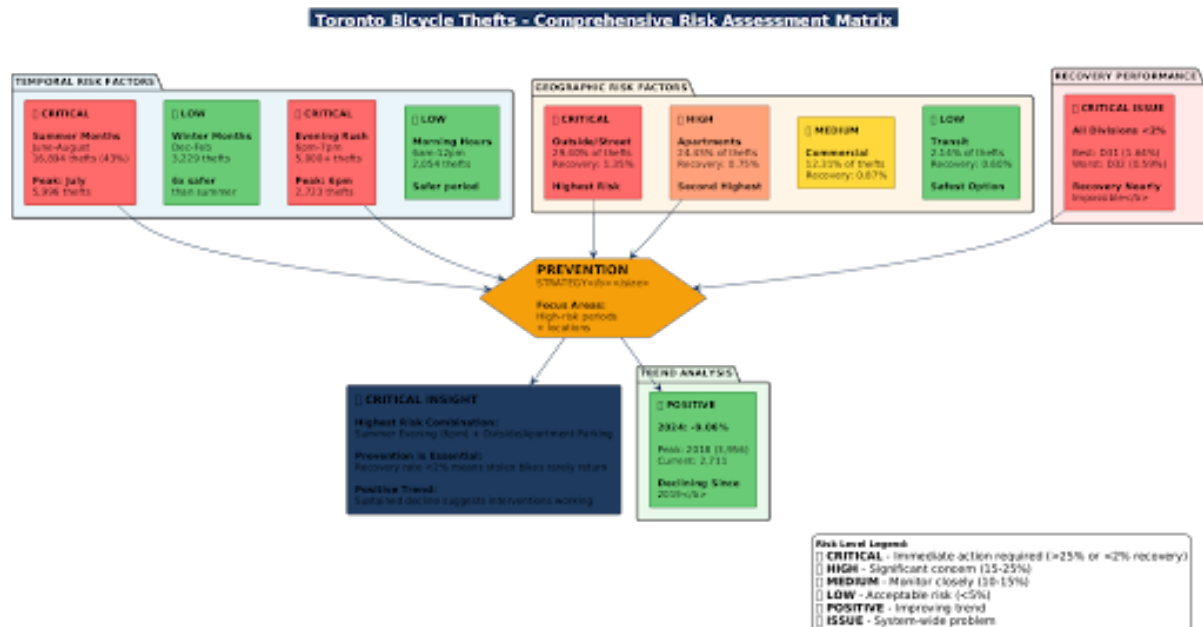


Figure 10: Multi-dimensional risk assessment matrix: temporal factors (summer, evening hours) and geographic factors (outside, apartments) converge with recovery crisis and positive trend analysis to inform prevention strategy

The matrix identifies critical risk factors ($\geq 25\%$ of thefts or $\leq 2\%$ recovery), high risk factors (15-25%), medium risk factors (10-15%), and low risk factors ($\leq 5\%$). This classification enables stakeholders to prioritize interventions on highest-impact areas while recognizing lower-priority concerns.

Critical Risk Combination: The highest-risk scenario combines summer months (June-August, 43% of annual thefts), evening hours (6pm-7pm rush, 5,000+ thefts), and outside/apartment locations (53.85% of spatial distribution). Cyclists parking bikes outdoors at apartments during summer evenings face compounded risk from multiple high-risk factors simultaneously.

Prevention Imperative: With recovery rates universally below 2%, the matrix emphasizes that prevention is not optional but essential. The systemic recovery failure means stakeholders must focus resources on theft deterrence rather than post-theft recovery efforts.

Positive Trajectory: Despite persistent challenges, the 2024 decline (-9.06%) continuing the multi-year downward trend from 2018 peak provides evidence that coordinated prevention strategies produce measurable results.

8 Strategic Recommendations

Based on comprehensive data analysis, we propose a four-pillar prevention strategy addressing all stakeholder groups and directly targeting the high-risk factors identified in the SQL analysis.

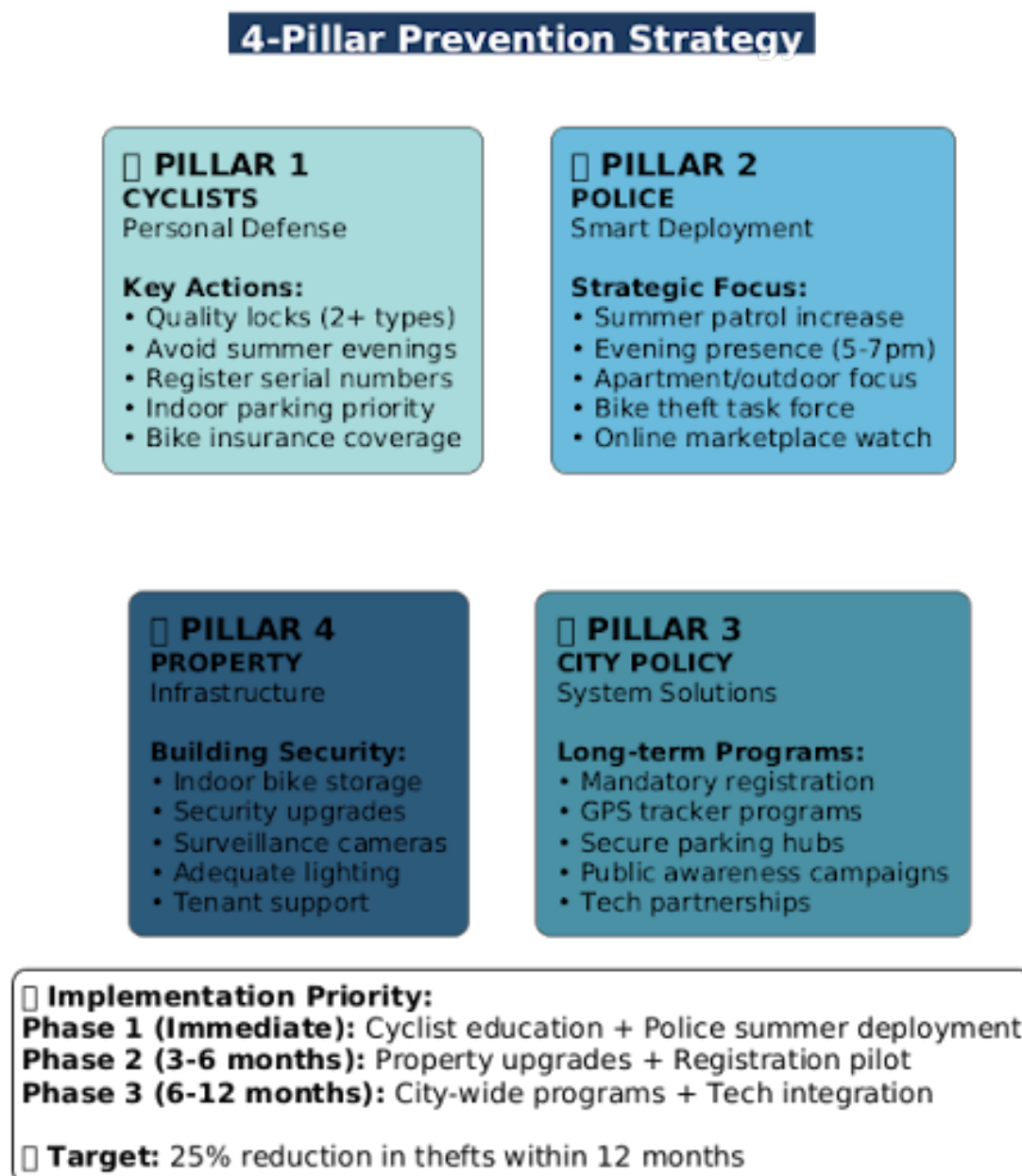


Figure 11: Four-pillar prevention strategy with phased implementation roadmap: coordinated approach addresses cyclists, police, city policy, and property infrastructure simultaneously

8.1 Pillar 1: Cyclist Personal Defense

Data-driven recommendations for individual cyclists include: quality lock investment (minimum two lock types - U-lock plus cable), temporal risk avoidance (minimize summer evening parking during 6-7pm rush hour), bike registration (serial number documentation addressing the 945 "UNKNOWN" bikes gap), location selection (prioritize indoor parking over outdoor/street options), and insurance coverage (given ~2% recovery likelihood).

8.2 Pillar 2: Police Smart Deployment

Strategic patrol allocation based on analytical findings: increase summer deployment (June-August accounts for 43% of annual thefts), evening presence enhancement (5-7pm coverage during highest-risk hours), geographic focus on apartments and outdoor areas (53.85% of theft locations), dedicated bike theft task force creation, and online marketplace monitoring (Craigslist, Facebook Marketplace surveillance for stolen bike resale).

8.3 Pillar 3: City Policy & Systems

Long-term structural interventions: mandatory bike registration program addressing the 2.4% "UNKNOWN" identification gap, GPS tracker subsidy programs improving recovery prospects, secure parking hub development in high-risk neighborhoods, public awareness campaigns timed for summer months, and technology partnerships with tracking solution providers.

8.4 Pillar 4: Property Infrastructure

Building-level security enhancements: indoor bike storage installation in apartment buildings (addressing 24.45% of thefts), security system upgrades (access control, monitoring), surveillance camera deployment in parking areas, lighting improvement programs, and tenant assistance with bike registration processes.

8.5 Implementation Roadmap

Phase 1 (Immediate): Cyclist education campaigns and police summer deployment - low-cost, high-impact interventions leveraging existing resources

Phase 2 (3-6 months): Property security upgrades and bike registration pilot program - infrastructure investments with measurable outcomes

Phase 3 (6-12 months): City-wide program rollout and technology integration - comprehensive system implementation with long-term sustainability

Success Target: Achieve 25% reduction in bicycle thefts within 12 months, building on the existing 9.06% annual decline trajectory. The target is evidence-based, considering that coordinated multi-stakeholder interventions can accelerate the positive trend already observed in the data.

9 Conclusion and Critical Insights

The deployment of the Toronto Bicycle Theft Risk Assessment Dashboard has enabled granular, data-driven evaluation of theft patterns across the Greater Toronto Area. By synthesizing over 39,000 records spanning 10 years through our automated cloud pipeline, we isolated critical insights defining the current risk landscape.

9.1 Macro-Trend Analysis: Positive Shift

Contrary to anecdotal perception, longitudinal SQL analysis reveals statistically significant decline in reported thefts. The 2024 year-to-date data indicates 9.06% decrease compared to 2023 (2,711 vs 2,981 thefts). This downward trajectory, sustained across multiple years from the

2018 peak (3,956 thefts), suggests that prevention interventions, infrastructure improvements, or behavioral changes are producing measurable impact.

Window function analysis precisely identified the growth period (2014-2018) characterized by the 2016 surge (+15.88%, +522 thefts) and the decline period (2019-2024) marked by the dramatic 2021 drop (-18.86%, -733 thefts, likely COVID-related). The multi-year consistency of the declining trend provides stronger evidence than single-year fluctuations, indicating structural rather than temporary changes in Toronto's bicycle theft landscape.

9.2 Operational Efficiency: The Recovery Crisis

Despite declining theft volume, recovery efficiency remains critically low, representing the most severe operational challenge identified. SQL analysis across all 17 Toronto police divisions reveals city-wide recovery rate consistently below 2%, with the best-performing division (D31) achieving merely 1.64% and worst performer (D32) at 0.59%.

This universally poor performance is independent of division characteristics - high-volume divisions (D52: 6,915 thefts, D51: 6,339 thefts) and low-volume divisions show comparable recovery failure rates. The systemic nature of this challenge indicates structural issues in bike recovery processes (lack of registration systems, difficulty tracking stolen bikes, challenges prosecuting theft cases) rather than division-specific operational deficiencies.

The recovery crisis fundamentally shifts prevention strategy priorities. With 98

9.3 Spatio-Temporal Risk Determinism

The risk profile exhibits high determinism, enabling predictive risk modeling. Outside/Street locations (29.40%) and apartments (24.45%) account for 53.85% of all thefts, indicating clear spatial vulnerability. Summer months (June-August) represent 43% of annual thefts with 6x multiplier versus winter months. The 6pm hour accounts for 2,723 thefts, precisely coinciding with evening commute when bikes are left unattended.

These patterns are consistent across the 10-year dataset, suggesting stable behavioral and environmental factors driving theft. The determinism enables actionable guidance: cyclists avoiding summer evening parking at apartments/outdoor locations can reduce personal risk exposure by over 50% based on the combined probability reduction from temporal and spatial factors.

9.4 Target Profiling: Universal Vulnerability

Brand analysis challenges assumptions about differential theft risk. While "OT" and "UK" generic categories account for 30.8% of thefts, premium brands (Specialized average \$1,880, Trek average \$1,545) are consistently targeted. The marginal recovery advantage for premium brands (Specialized: 2.97% vs UK: 0.26%) provides insufficient justification for reduced security measures on budget bikes.

The 945 bikes (2.4%) labeled "UNKNOWN MAKE" represent a critical data quality and policy gap. Without manufacturer identification, police cannot match recovered bikes to theft reports, rendering recovery impossible even when bikes are physically located. This finding strongly supports mandatory bike registration systems as a policy intervention.

9.5 Summary

Bicycle theft in Toronto is highly seasonal, location-specific, and brand-agnostic with near-zero recovery rates. The comprehensive analysis identified four critical insights:

1. **Positive trend:** 9.06% decline in 2024 continuing sustained improvement since 2019, evidencing intervention effectiveness
2. **Critical risk combination:** Summer evenings (6-7pm) at apartments and outdoor locations represent highest-probability theft scenarios
3. **System-wide recovery challenge:** 12% recovery rate across all divisions necessitates prevention-focused rather than recovery-focused strategies
4. **Actionable intelligence:** Clear deterministic patterns enable targeted interventions for cyclists, police, property managers, and city planners

The cloud pipeline successfully transforms historical police data into predictive analytical tools, empowering stakeholders to shift from reactive incident reporting to proactive risk avoidance strategies. The data products developed - 15 analytical queries, 6 SQL views, master table architecture, and interactive dashboard - provide robust analytical infrastructure for continued monitoring, policy evaluation, and strategic decision-making in urban bicycle theft prevention.

10 References

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